Dataset shift in neural models

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Prepare the purpose, prerequisites and materials

> Prepare Dataset shift in neural models: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Explain how a changed **input** mix can change overall accuracy

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Uncertainty and failure analysis

Without opening the old lesson, explain one key idea from Uncertainty and failure analysis. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a predicted class, confidence score and observed correctness

Reference example, not the only acceptable wording: Teacher explanation: B predicts 1 with confidence 0.90 and is wrong. E predicts 0 with confidence 0.55 and is correct. F predicts 0 with confidence 0.90 and is wrong.

F's low score for class 1 corresponds to a high score for class 0, the class selected.

Similarity and normalisation

Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate vector length and normalise a nonzero vector

Reference example, not the only acceptable wording: Teacher explanation: Lengths are 5, 10 and 5. Unit vectors are [0.6, 0.8], [0.6, 0.8] and [-0.6, -0.8].

The length difference between Q and A disappears; the opposite direction of C is preserved.

Augmentation and leakage

Without opening the old lesson, explain one key idea from Augmentation and leakage. Give an example and a mistake that the idea helps you avoid.

Look for: Check whether a transformation preserves the task label

Reference example, not the only acceptable wording: Teacher explanation: Original d = -2 and class 0. Row swapping leaves d = -2 and class 0. Column swapping gives rows [1, 0], [1, 0], d = 2 and class 1.

Use two 45-minute sessions. Work from counts before percentages and keep the frozen model separate from the changing sample mix.

The exact equal group rates are constructed for arithmetic; they do not establish a population law. A **monitoring** alert should lead to **evidence** gathering rather than an automatic causal claim or re**training** decision.

Outcomes

Explain how a changed **input** mix can change overall accuracy

Calculate group and overall results with the correct denominators

Plan a shift investigation while preserving independent evaluation

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

No. Confidence is a model **output**, not a guarantee of familiarity or correctness. Check **input** conditions and independently verified outcomes rather than treating confidence as proof.

End goal for the pictorial case

By the end, I can explain **dataset shift** in neural models using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain and model the mechanism

> Model Dataset shift in neural models using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Explain Dataset shift in neural models · 1

A neural image classifier developed using mostly clear pictures may later receive mostly dim pictures. Its learned weights can stay unchanged while its overall results change.

Dataset shift concerns changes in the data encountered, including **input**s, label frequencies or the relationship between **input**s and labels.

A shift does not always reduce accuracy, and an accuracy change alone does not identify its cause.

Explain Dataset shift in neural models · 2

Use the fictional audit below. Clear and dim are disjoint groups defined before the audit, and every case belongs to exactly one group. The same model checkpoint, pre**process**ing and **prediction** threshold are used throughout.

Counts are constructed teaching data, not results from a deployed model. Both cohorts have checked reference labels.

Clear accuracy is 72 / 80 = 90 percent in development and 18 / 20 = 90 percent in new use. Dim accuracy is 10 / 20 = 50 percent and 40 / 80 = 50 percent.

The observed group rates are unchanged, but the group proportions are different.

Explain Dataset shift in neural models · 3

Development accuracy is (72 + 10) / (80 + 20) = 82 percent. New-use accuracy is (18 + 40) / (20 + 80) = 58 percent. The drop is 24 percentage points.

Taking the unweighted mean of 90 and 50 gives 70 percent, which is not either cohort's overall accuracy because neither cohort has equal group sizes.

Explain Dataset shift in neural models · 4

Each pair is (cases, correct), and the function assumes integer counts for non-overlapping groups. It rejects empty **input**s, groups with no cases and impossible correct counts.

A group with no observed cases has an undefined empirical accuracy, not zero accuracy. Do not count the same example in two groups when forming an overall total.

Equivalently, weight each group accuracy by its share of cases: development is 0.80 times 0.90 + 0.20 times 0.50 = 0.82; new use is 0.20 times 0.90 + 0.80 times 0.50 = 0.58.

Term	Meaning
Dataset shift	A change in the distribution of inputs, labels or their relationship between settings
Subgroup	A declared subset of examples sharing a relevant condition
Group mix	The proportions of examples belonging to different groups
Monitoring	Repeatedly checking recorded inputs, outputs and available outcomes against a reference

The same group rates meet a different mix

GROUP	DEVELOPMENT SHARE	NEW-USE SHARE
Clear, 90% correct	80 of 100	20 of 100
Dim, 50% correct	20 of 100	80 of 100

The weaker group becomes more common. Overall mistakes rise from 18 of 100 to 42 of 100 without any change in these observed within-group accuracy rates.

Explain Dataset shift in neural models · 5

Guided: 45 clear plus 25 dim cases are correct, giving 70 of 100 or 70 percent. This assumes exactly the stated group rates hold in that cohort. Exit: the mix change is observed, but accuracy is unknown without checked labels.

Gather a labelled audit and check model/pipeline versions. Cases used to choose or fit a change cannot also provide independent final **evidence**.

Reserve new appropriate cases and compare original and revised models on the same cases with group and overall results.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

The same group rates meet a different mix

GROUP	DEVELOPMENT SHARE	NEW-USE SHARE
Clear, 90% correct	80 of 100	20 of 100
Dim, 50% correct	20 of 100	80 of 100

The weaker group becomes more common. Overall mistakes rise from 18 of 100 to 42 of 100 without any change in these observed within-group accuracy rates.

Model the reasoning aloud · 6

The arithmetic explains this constructed aggregate difference. It does not prove that dim lighting caused each real error or that these group rates will stay fixed in future samples.

Report group counts and uncertainty about generalisation. Review specific errors and reference labels as in Week 29.

If a new cohort has 50 clear and 50 dim cases, and exactly 90 percent and 50 percent are correct respectively, how many correct cases and what overall accuracy result? State the assumption behind this calculation.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Dataset shift in neural models

Separate within-setting performance from changing setting proportions.

The facts we will use

Supplied facts

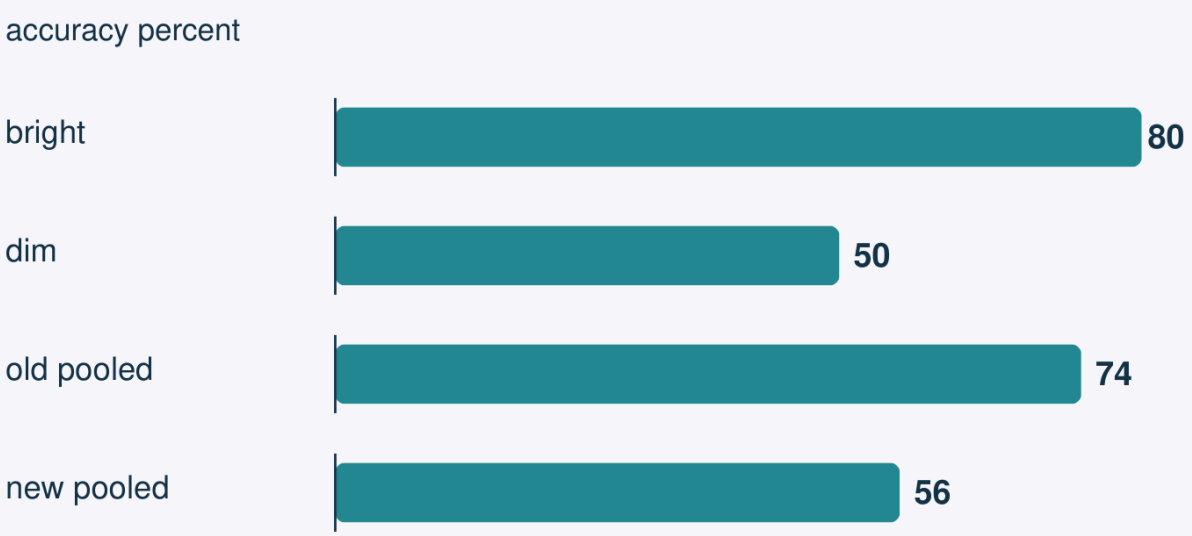
Toy old-use mix: bright 40 cases32 correct; dim 10 cases5 correct. New-use mix: bright 10 cases8 correct; dim 40 cases20 correct.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/30

Fixed slice rates, changed pooled accuracy



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/30

Dataset shift in neural models

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep slice rates

Bright is 80% in both mixes; dim is 50% in both.



2. Old pooled rate

Old accuracy 37/50=74%.



3. New pooled rate

New accuracy 28/50=56%; more use now occurs in the harder dim setting.

CHANGE ONE THING

New-use mix returns to40 bright and 10 dim with the same slice rates.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Keep slice rates	Bright is 80% in both mixes; dim is 50% in both.
2. Old pooled rate	Old accuracy 37/50=74%.
3. New pooled rate	New accuracy 28/50=56%; more use now occurs in the harder dim setting.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A pooled rate depends on how often each condition appears as well as performance within that condition. Bright and dim accuracy stay fixed in this example. The new use contains more dim cases, so its pooled accuracy falls.

Report both group counts and group rates before concluding that the network itself deteriorated.

What this example does not establish

Keep the stated limit

Real shift can also change labels, sensors and within-slice behaviour; this toy decomposition

isolates only mixture change.

Change one thing in the pictured case

Change one thing

Change: New-use mix returns to40 bright and 10 dim with the same slice rates.

Predicted result

Prediction: Pooled accuracy returns to74%.

Why the result follows

Reason: The mixture weights change, while each slice's conditional performance stays fixed.

Ask while showing the picture

Ask while showing the picture · 7

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Dataset shift in neural models, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve Week 29's warning that confident outputs can be wrong on unfamiliar inputs.
7 to 18 minutes	Read the cohort table and complete W1. Mark fixed checkpoint and pipeline versus changed mix.
18 to 32 minutes	Compute totals and W2. Compare the weighted result with the tempting unweighted average.
32 to 45 minutes	Trace the optional count function and the visual; explain why group sizes affect overall accuracy.
Session 2 start to 8 minutes	Complete the guided balanced cohort and name the fixed-rate assumption.
8 to 20 minutes	Complete W3 and discuss missing labels, pipeline changes and possible learned shortcuts.
20 to 35 minutes	Collect W4 independently, requiring counts, group rates, both totals and percentage points.
35 to 45 minutes	Use W5 and the exit response to protect independent evaluation. Give feedback, then a fresh retry.

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide separate columns for cases, correct, errors and group share. Use 100 counters or a drawn 100-cell grid to change the mix while keeping group rates fixed.

Extension: with clear share s and fixed rates 0.90 and 0.50, overall accuracy is 0.50 + 0.40*s. Test s = 0, 0.5 and 1, and explain why observed data need not keep those rates fixed.

Reduce help deliberately

Completion practice

Completion task: Calculate overall correct counts, accuracy and errors in both cohorts, plus the percentage-point drop. Explain why averaging 90 percent and 50 percent without weights gives the wrong overall result here.

Reasoning to listen for

Teacher check: Teacher explanation: Development has 82 correct and 18 errors out of 100, or 82 percent accuracy. New use has 58 correct and 42 errors out of 100, or 58 percent. The drop is 24 percentage points.

An unweighted mean gives 70 percent and ignores the unequal group sizes.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Dataset shift** in neural models. Proposed explanation: “Average the percentages regardless of group sizes”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Overall accuracy is total correct divided by total cases; group accuracies need weights proportional to group size.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Did the supplied model’s dim-light accuracy deteriorate? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

No, it remains50%. The pooled drop from 74% to56% reflects a larger proportion of dim cases, not changed within-slice rates. Changed-case check: Pooled accuracy returns to74%.

The mixture weights change, while each slice's conditional performance stays fixed.

Changed-case reasoning: Pooled accuracy returns to74%. The mixture weights change, while each slice's conditional performance stays fixed.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the investigation and diagnose misconceptions

> Run the Dataset shift in neural models investigation and use the first unsupported step to choose a teaching response.

Average the percentages regardless of group sizes: Overall accuracy is total correct divided by total cases; group accuracies need weights proportional to group size.

A shift alert proves a loss of accuracy: **Input** changes may help, hurt or leave accuracy unchanged; verified outcomes are needed to measure performance.

An unchanged **input** summary proves nothing changed: A summary can miss relevant differences, and the **input**-label relationship or labelling policy can change without an obvious **input** shift.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Dataset shift in neural models questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Guided: 45 clear plus 25 dim cases are correct, giving 70 of 100 or 70 percent. This assumes exactly the stated group rates hold in that cohort. Exit: the mix change is observed, but accuracy is unknown without checked labels.

Gather a labelled audit and check model/pipeline versions. Cases used to choose or fit a change cannot also provide independent final **evidence**.

Reserve new appropriate cases and compare original and revised models on the same cases with group and overall results.

W1 Audit the two cohorts

Workbook question · W1

From the table, calculate clear and dim accuracy in each cohort and the clear/dim proportions. Which observed quantities change, and which remain the same?

Teacher reasoning · W1

Teacher explanation: Clear accuracy is 90 percent in both cohorts; dim accuracy is 50 percent in both. Development has 80 percent clear and 20 percent dim; new use has 20 percent clear and 80 percent dim.

The mix changes while observed group rates remain the same.

W2 Use the right denominator

Workbook question · W2

Calculate overall correct counts, accuracy and errors in both cohorts, plus the percentage-point drop. Explain why averaging 90 percent and 50 percent without weights gives the wrong overall result here.

Teacher reasoning · W2

Teacher explanation: Development has 82 correct and 18 errors out of 100, or 82 percent accuracy. New use has 58 correct and 42 errors out of 100, or 58 percent. The drop is 24 percentage points.

An unweighted mean gives 70 percent and ignores the unequal group sizes.

W3 Separate an alert from an outcome

Workbook question · W3

New-use **monitoring** shows more dim images but reference labels have not arrived. What can you report now, what can you not calculate, and what checks should precede blaming the neural network?

Teacher reasoning · W3

Teacher explanation: Report the observed change in group proportions with counts and period. Current accuracy and error patterns need verified outcomes.

Check checkpoint and pre**process**ing versions, group definitions, **input** quality and label policy; collect a representative labelled audit sample before concluding how performance changed.

W4 Analyse a new mixture independently

Workbook question · W4

A separate audit has development groups P: 60 cases, 48 correct and Q: 40 cases, 20 correct. New-use groups are P: 20 cases, 16 correct and Q: 80 cases, 40 correct.

Compute each group rate, both overall accuracies, and the percentage-point change. Under the same group rates, what accuracy would equal-sized P and Q groups have?

Teacher reasoning · W4

Teacher explanation: P is 80 percent correct and Q is 50 percent correct in both cohorts. Development accuracy is 68 of 100, or 68 percent; new-use accuracy is 56 of 100, or 56 percent. The drop is 12 percentage points.

Equal-sized groups at those rates give 65 percent accuracy.

W5 Protect the comparison

Workbook question · W5

A team retrains using new-use errors, then evaluates on those same images and their brightened copies. Explain why the score is not independent **evidence** and propose a comparison that includes the original model.

Teacher reasoning · W5

Teacher explanation: The evaluated cases influenced **training**, and transformed copies share source content. Keep related images in one split and reserve untouched source groups or an appropriate later period.

Evaluate original and revised models on identical reserved examples, with checked labels, group counts and rates, overall accuracy, and documented pre**process**ing.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4 requires group rates 80 and 50 percent, overall accuracies 68 and 56 percent, a 12-percentage-point drop, and equal-group accuracy 65 percent.

Fresh retry: development P has 75 cases with 60 correct; Q has 25 with 10 correct. New use P has 25 with 20 correct; Q has 75 with 30 correct. Group rates are 80 and 40 percent in both.

Overall accuracy changes from 70 to 50 percent, a 20-percentage-point drop. Equal-sized groups at those rates give 60 percent.

Assess independent transfer

Independent question · withhold the key

Selected case: A separate audit has development groups P: 60 cases, 48 correct and Q: 40 cases, 20 correct. New-use groups are P: 20 cases, 16 correct and Q: 80 cases, 40 correct.

Compute each group rate, both overall accuracies, and the percentage-point change. Under the same group rates, what accuracy would equal-sized P and Q groups have?

Selected case answer

Teacher key · Selected case answer

Teacher explanation: P is 80 percent correct and Q is 50 percent correct in both cohorts. Development accuracy is 68 of 100, or 68 percent; new-use accuracy is 56 of 100, or 56 percent. The drop is 12 percentage points.

Equal-sized groups at those rates give 65 percent accuracy.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect the project, support and next lesson

> Connect Dataset shift in neural models to Image-pattern Network Investigation, adapt support and plan a later changed-case check.

Add a labelled **dataset shift** in neural models **evidence** sheet to Image-pattern Network Investigation. Show how you can explain how a changed **input** mix can change overall accuracy.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide separate columns for cases, correct, errors and group share. Use 100 counters or a drawn 100-cell grid to change the mix while keeping group rates fixed.

Extension: with clear share s and fixed rates 0.90 and 0.50, overall accuracy is 0.50 + 0.40*s. Test s = 0, 0.5 and 1, and explain why observed data need not keep those rates fixed.

Week 31 measures efficiency and resource use while keeping task quality, hardware and comparison conditions explicit.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/31, ai-5/33, ai-5/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Dataset shift in neural models lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Explain how a changed **input** mix can change overall accuracy

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Calculate overall correct counts, accuracy and errors in both cohorts, plus the percentage-point drop.

Explain why averaging 90 percent and 50 percent without weights gives the wrong overall result here.

Supported practice answer

Reasoned practice answer: Teacher explanation: Development has 82 correct and 18 errors out of 100, or 82 percent accuracy. New use has 58 correct and 42 errors out of 100, or 58 percent. The drop is 24 percentage points.

An unweighted mean gives 70 percent and ignores the unequal group sizes.

Check every learner before discussion: Use the worked case for **Dataset shift** in neural models. Proposed explanation: “Average the percentages regardless of group sizes”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Overall accuracy is total correct divided by total cases; group accuracies need weights proportional to group size.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: New-use mix returns to40 bright and 10 dim with the same slice rates. Require a **prediction** and a reason before comparing. Expected result: Pooled accuracy returns to74%.

Reason: The mixture weights change, while each slice's conditional performance stays fixed.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

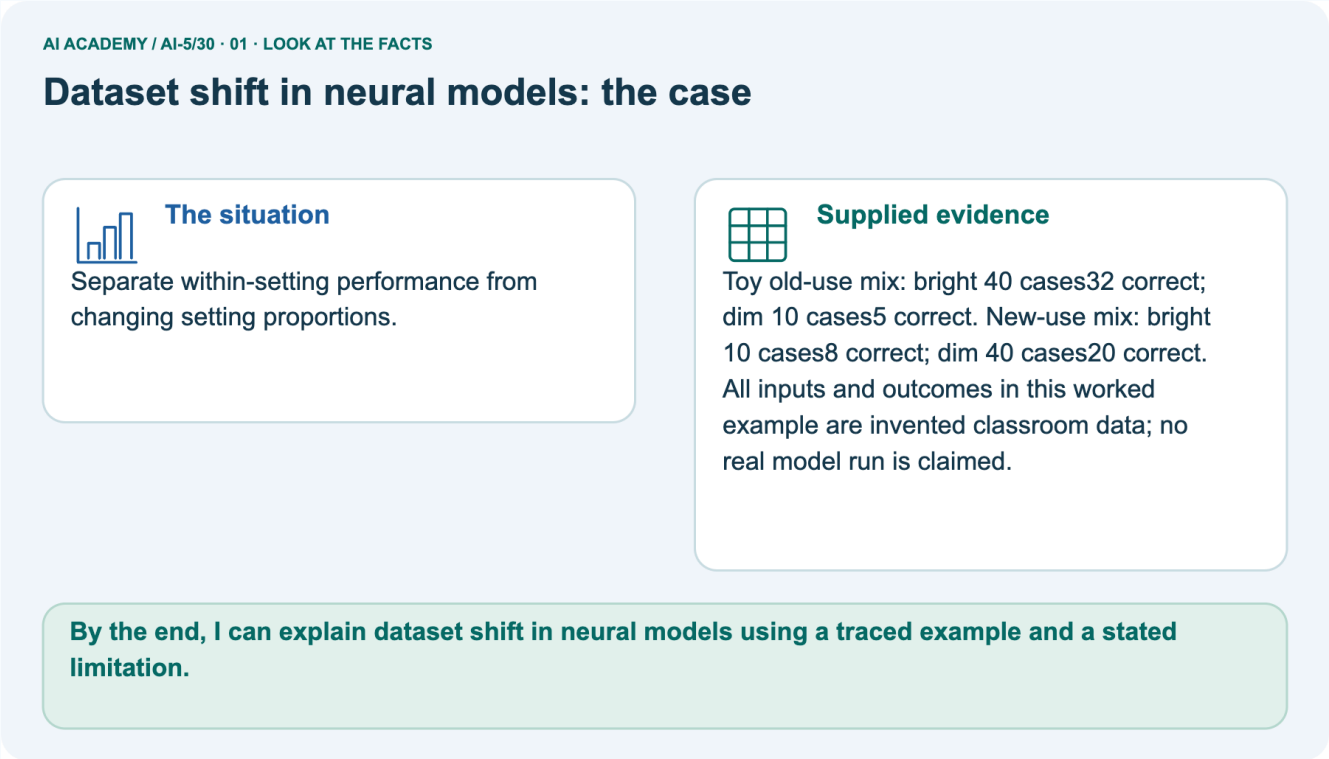
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Dataset shift in neural models graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Dataset shift in neural models: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Separate within-setting performance from changing setting proportions. Supplied facts: Toy old-use mix: bright 40 cases32 correct; dim 10 cases5 correct. New-use mix: bright 10 cases8 correct; dim 40 cases20 correct.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

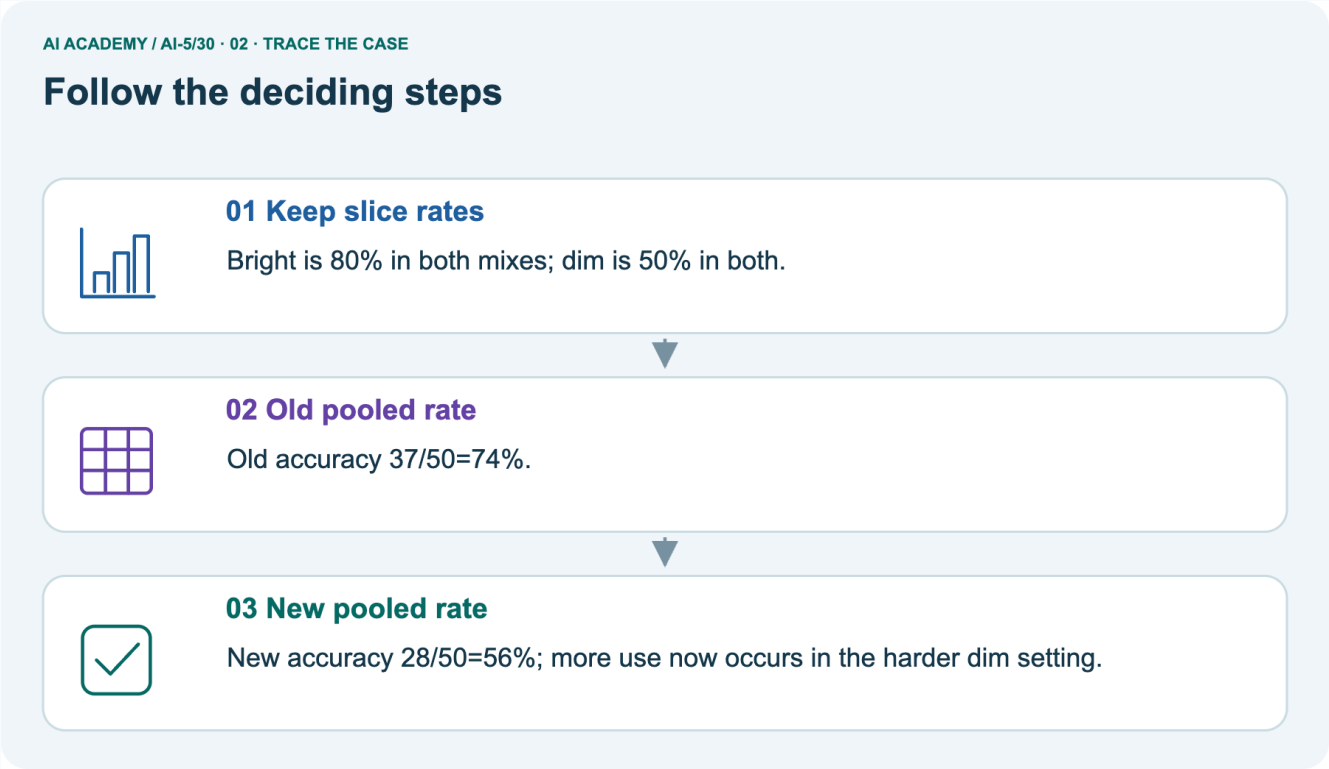
Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the

image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep slice rates: Bright is **80%** in both mixes; dim is 50% in both. → Old pooled rate: Old accuracy 37/50=74%. → New pooled rate: New accuracy 28/50=56%; more use now occurs in the harder dim setting.

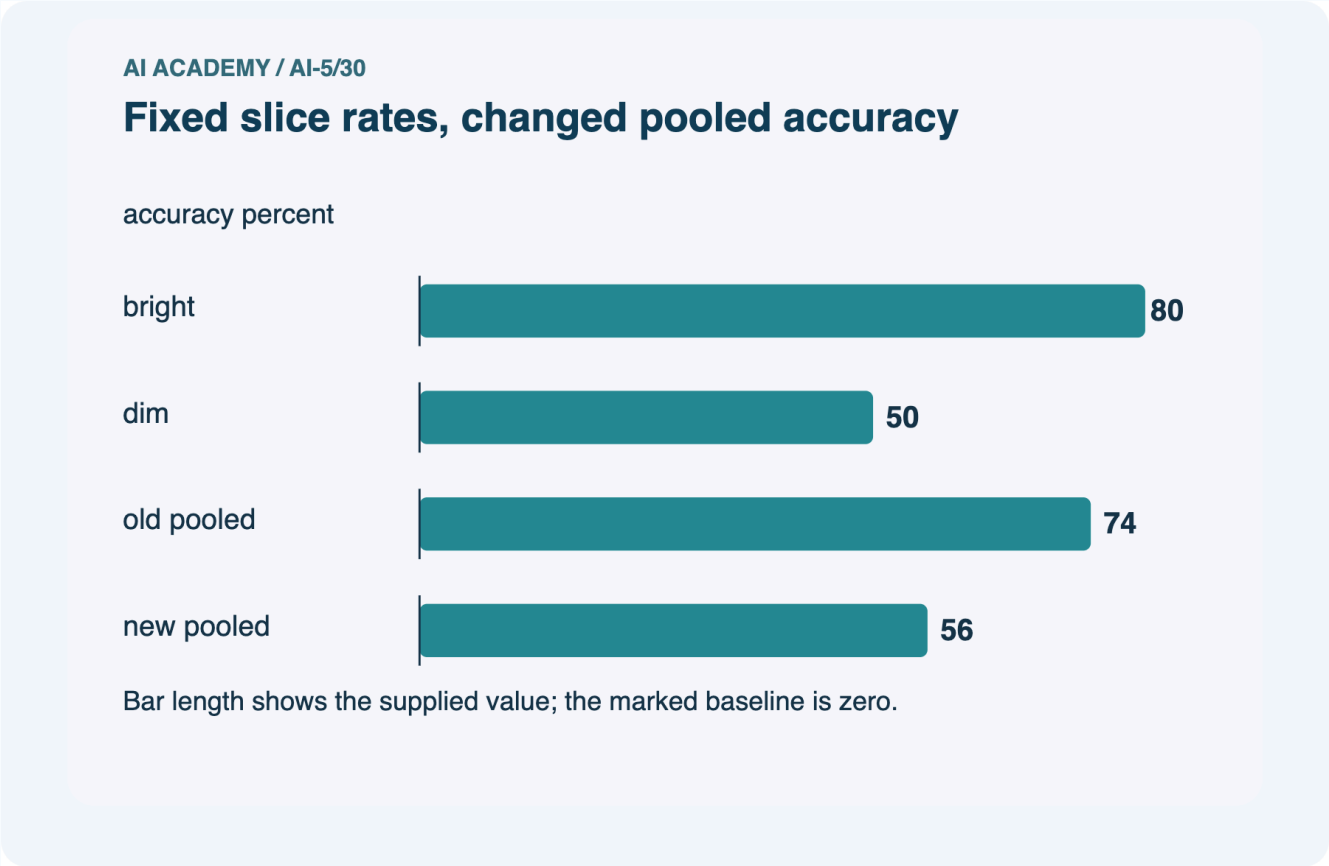
Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if

the image is difficult to see.

03. Fixed slice rates, changed pooled accuracy



Follow the labelled picture with the matching explanation. Open the vector diagram

Fixed slice rates, changed pooled accuracy. Units: accuracy percent. bright: 80. dim: 50. old pooled: 74. new pooled: 56.

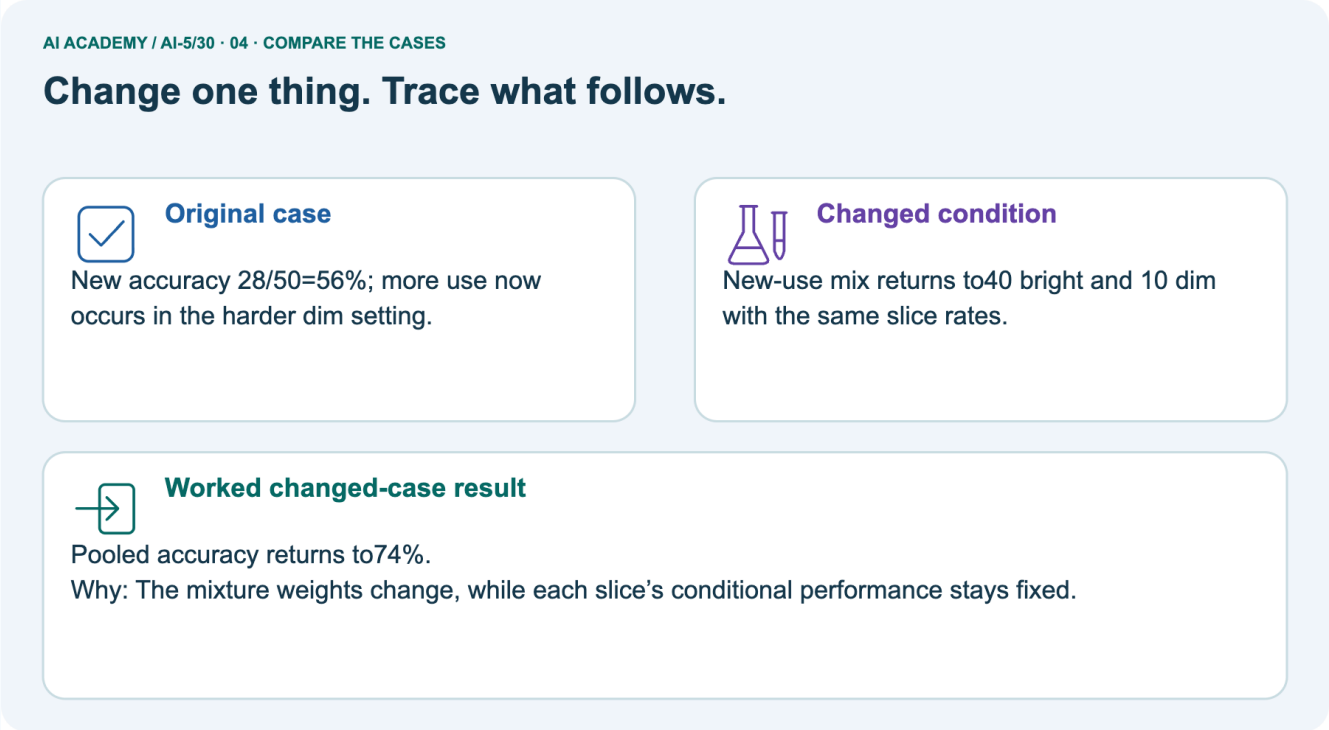
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: New accuracy 28/50=56%; more use now occurs in the harder dim setting. Changed condition: New-use mix returns to40 bright and 10 dim with the same slice rates. Expected result: Pooled accuracy returns to74%.

Reason: The mixture weights change, while each slice's conditional performance stays fixed.

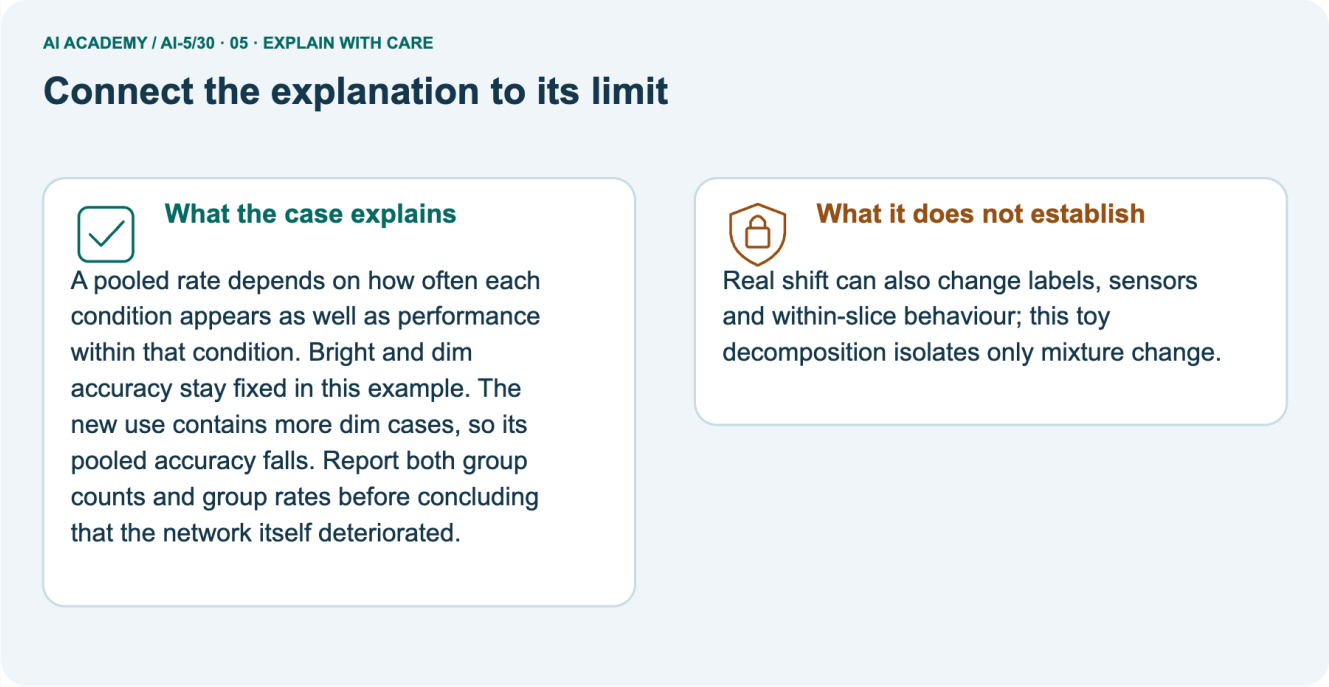
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The mixture weights change, while each slice's conditional performance stays fixed.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A pooled rate depends on how often each condition appears as well as performance within that condition. Bright and dim accuracy stay fixed in this example. The new use contains more dim cases, so its pooled accuracy falls.

Report both group counts and group rates before concluding that the network itself deteriorated.

Limit: Real shift can also change labels, sensors and within-slice behaviour; this toy decomposition isolates only mixture change.

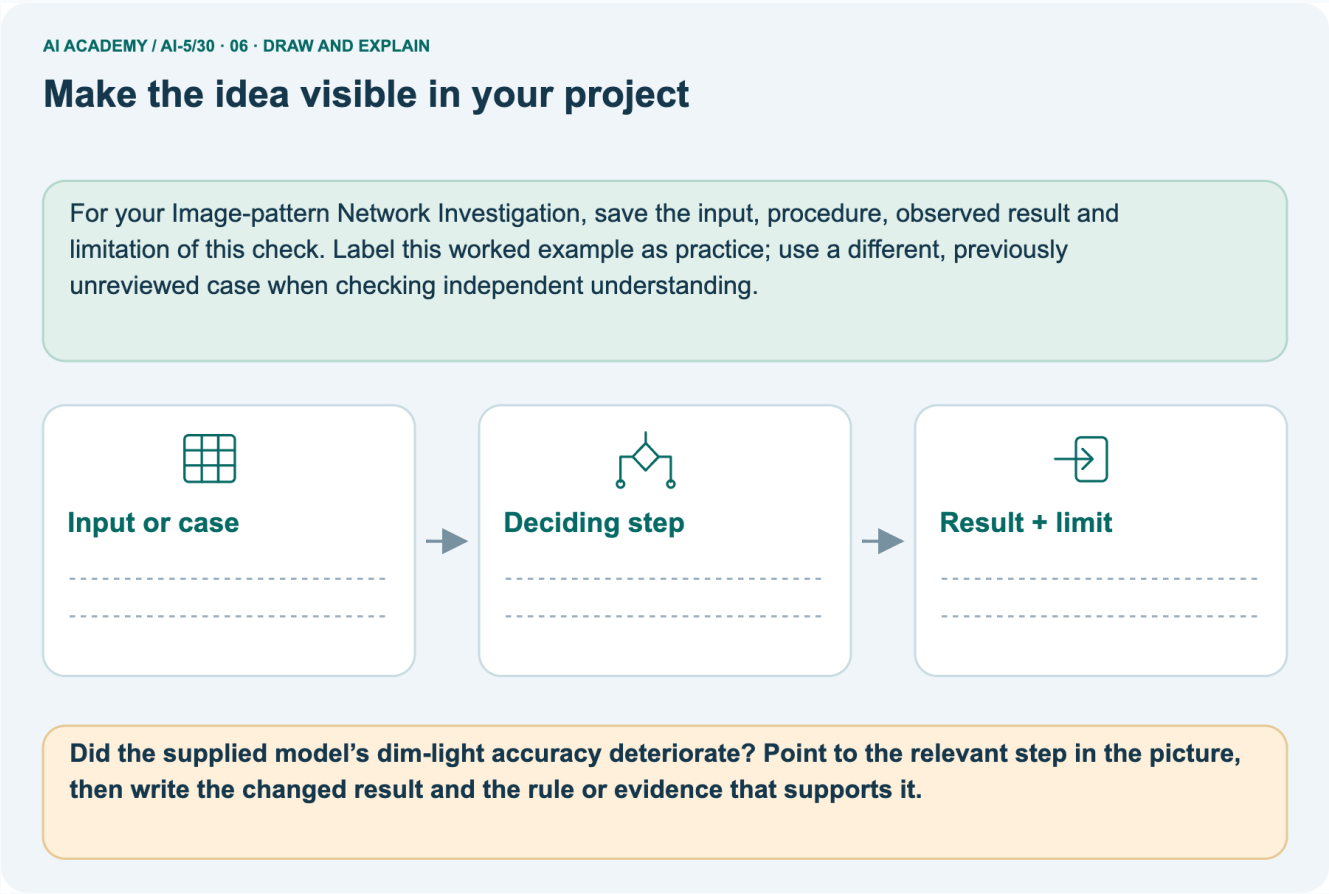
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Real shift can also change labels, sensors and within-slice behaviour; this toy decomposition isolates only mixture change.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Did the supplied model's dim-light accuracy deteriorate? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No, it remains50%. The pooled drop from 74% to56% reflects a larger proportion of dim cases, not changed within-slice rates. Changed-case check: Pooled accuracy returns to74%.

The mixture weights change, while each slice’s conditional performance stays fixed.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.